Interim report: Phase 2

Intelligent Agriculture Information System, Andaman & Nicobar Islands
CS F/U 407 Artificial Intelligence · Team 34 · Phase 2 (data assimilation, working agents, database)

1. Summary

Phase 2 turns the Phase 1 corpus into a provenance-tracked knowledge base using an agentic pipeline.

- **Corpus:** 20 government files from DES A&N, Dept. of Agriculture, NRSC/Bhuvan and MoFPI.
- **Agents:** seven specialist extraction agents behind a router, sharing a normalizer and a validator, orchestrated with LangGraph.
- **Model:** a local Qwen2.5-3B model with a project LoRA adapter does the language work: reading prose profiles, interpreting unknown table headers, resolving unseen names, and routing ambiguous files.
- **Code:** handles bulk cell extraction, checks and provenance.

Headline results (production batch `B20261016_035234` + island re-run `B20261016_041718`):

- **Coverage:**
  - 20/20 files reached a terminal status with **0 manual interventions** (automation rate 1.0).
  - 15 files loaded; 4 bilingual files and D07 are partial (some columns unmapped); D18 is pending; D19/D20 are skipped by design.
  - **32,167** numeric facts and **74** evidence-backed text facts are in the database.
- **Accuracy:**
  - **Groundedness 100%:** every loaded number's printed text is found on its cited page.
  - **48/48** values read by eye from rendered pages match the database.
  - Routing is **20/20**.
- **LoRA adapter vs base model (held-out documents):**
  - island extraction F1 **0.27** → **0.94**
  - column mapping **0.03** → **0.64**
  - on a reworded, unseen layout, raw F1 **0.35** → **0.72**, where the regex baseline drops to 0.61.

2. Problem framing (from Phase 1, unchanged)

The goal is to answer spatial, temporal and provenance questions about A&N agriculture (production, prices, inputs, land use, holdings, schemes) from official sources, every answer traceable to a page and cell. Phase 2 builds and measures the extraction layer and the database; Phase 3 adds the query agent.

3. Sources and manifest

- `data/manifest.csv` records, per file:
  - source and portal
  - download URL from the Phase 1 inventory (DES ecostat portal, Bhuvan LULC viewer, MoFPI state profile)
  - collection date

    - languages and scripts (Kruti Dev detected automatically)
    - modality, pages, SHA-256
    - licence

- Licence position: DES and Agriculture Department portals state no licence, so their files are flagged *academic use only*. The MoFPI report is "all rights reserved", so it is used for academic extraction only and not redistributed. NRSC/Bhuvan publications may be used non-commercially with attribution.
- No personal data is extracted. The island profiles and machinery tables are aggregate counts; no farmer-level records exist in the corpus.

4. Sources → agents

The full map and agent cards are in source_agent_map.md. In brief:

Group	Files	Agent	Hard part
G1 matrix tables	D01, D02, D07	matrix_table_agent	merged/rotated headers, zone names, footnote units
G2 bilingual tables	D03–D06	bilingual_table_agent	Hindi in Kruti Dev legacy font, 13 layouts per file
G3 price tables	D11–D13	price_table_agent	rotated headers, raster ruling (2025), misprinted year rows
G4 island profiles	D08–D10	island_profile_agent (LLM)	interleaved prose, synonyms, NIL values
G5 narrative	D14, D15	narrative_agent (LLM facts)	segmentation, grounding of free text
G6 screenshot	D16	image_table_agent	OCR, merged/dropped cells
G7 LULC	D17 (D18 pending)	lulc_agent	class hierarchy, other UTs
G8 raster maps	D19, D20	skip (router)	out of scope by brief

5. Challenge and literature

See challenge_literature.md. The focus is layout and encoding traps in government PDFs, above all Hindi set in the Kruti Dev legacy font. No general tool we tested (pdfplumber, PyMuPDF, Docling) recovers it; our glyph-map decoder recovers 8/8 test words. Raster-ruled tables, rotated headers and OCR damage are handled by targeted tools and measured against values read from the pages.

6. Orchestration

Supervisor/router + per-document pipeline (route → extract → validate → retry? → load) in LangGraph. Details, alternatives and failure handling: orchestration.md.

7. Evaluation and automation

Full tables are in evaluation.md, generated from `eval/results/`.

Area	Result	Target

Direct value checks (hand-read)	48/48	≥ 0.95 ✓
Groundedness of loaded numbers	1.00 (all agents)	≥ 0.99 ✓
Routing (deployed hybrid)	20/20 on corpus; 0.95 when the model is asked on ambiguous files	≥ 0.8 ✓
Island extraction F1 (adapter, gated)	0.935 (base 0.265)	≥ 0.8 ✓
Column mapping (adapter)	0.635 (base 0.027)	≥ 0.8 ✗, mitigated by id checks
JSON validity, island (adapter)	0.61 strict, rest salvaged	≥ 0.95 ✗, known loop issue
Automation (inputs without manual intervention / attempted)	20/20	≥ 0.9 ✓
LLM calls in the production run	531; 157 model outputs rejected by gates	—

- **Evaluators used (Foundry menu):** tool selection, trajectory (AgentEvals superset match), tool input accuracy, task completion, task adherence, groundedness, navigation efficiency.
- **Skipped, with reasons:** intent resolution, relevance/coherence/fluency and safety. They need natural-language answers, which arrive in Phase 3.
- **LangSmith:** held-out sets uploaded as datasets; base vs adapter recorded as 4 experiments; a 5-document pipeline batch traced end to end (project). Links in evaluation.md.

8. Database

SQLite star schema (fact table `observation` + lookup tables + provenance). The team's Production / Price / Source design is preserved as `v_production`, `v_price`, `source`/`document`. Schema, identifiers, missing values, duplicates and validation rules are in schema.md. The 10 sample queries with expected results are in sample_queries.md: spatial (Q1, Q2, Q6), temporal (Q3, Q4, Q8), provenance (Q5), cross-source (Q7), census (Q9) and land use (Q10).

The database (`db/34_AN_knowledgebase_v1.sqlite`, 14.6 MB) holds:

What	Count
Observations	32,167, of which 2,264 are explicit missing values with a reason and 645 carry a validation warning
Commodities	148
Locations	45 (UT, districts, tehsils, zones, islands, market)
Metrics	39
Periods	2005-06 to 2025-26
Text facts	74
Cross-source disagreements (`v_conflict`)	341
Hindi crop names	40, filled automatically from decoded Kruti Dev labels

9. Team 34

Name	ID

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10. Pending for Phase 3

1. **D18 atlas:** chart digitisation, or source the underlying NRSC state tables.
2. **D15 memoranda:** turn the subsidy line items (quantity × amount) into observations.
3. **Gazetteer:** add LGD codes to the location table.
4. **Instructor files:** widen layout coverage (more unseen price/table layouts) and retrain the adapter with any instructor files allowed into training.
5. **Query agent:** natural-language → SQL with provenance, including the 10-query acceptance test (2 spatial, 2 temporal, 1 provenance), using `document_chunk` for retrieval.
6. **Hindi names:** a few decoded names carry spacing artefacts from the PDF text layer (e.g. सरसां े), and one DES row prints तिलहन (oilseeds) next to "Seasam". Needs a Unicode normalisation pass plus manual review.